

GAUD-ECollective Network: A Research Architecture for a Globally Distributed and Collectively Governed Artificial Intelligence

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Abstract—Modern artificial intelligence is constrained by concentrated access to compute, training data, model expertise, and capital. This paper proposes the GAUD-E Collective Network, an open protocol for building and operating a shared global artificial intelligence through a distributed community of people, organizations, and computing devices. The central design principle is collective intelligence rather than per-user personalization: participants contribute computation, evaluation, software, research, storage, or other explicitly authorized resources to improve a common model ecosystem. The architecture combines federated and low-communication learning, peer-to-peer networking, cryptographic contribution receipts, independent verification, and an EVM-compatible blockchain settlement layer. Blockchain is deliberately kept out of the high-volume tensor path; it records identities, commitments, proofs, governance events, and economic settlements while model training and inference occur off-chain. The paper defines a proposed Proof-of-AI-Contribution (PoAIC) research direction, an incentive model based on verified value rather than raw hardware ownership, and a privacy architecture in which user data remains local by default and only consented model updates or aggregates leave the device. Related systems such as Federated Learning, DiLoCo, Gensyn, Prime Intellect, Bittensor, and decentralized compute marketplaces demonstrate that several individual components are already feasible. The contribution of this paper is a systems-level blueprint for integrating these components into a single collective-AI protocol, together with a rigorous evaluation agenda covering convergence, verification cost, economics, security, privacy, and environmental impact. The paper makes no claim that decentralization automatically reduces cost or emissions; these are hypotheses that require empirical measurement.

Index Terms—collective artificial intelligence, federated learning, distributed training, blockchain, decentralized AI, proof of useful work, verifiable machine learning, peer-to-peer computing, AI economics, energy efficiency.

I. INTRODUCTION

The scaling of modern AI has created a concentration problem. Advanced training and inference increasingly depend on specialized accelerators, large data-center infrastructure, high-bandwidth interconnects, and capital-intensive engineering. The International Energy Agency (IEA) reports that data-center electricity consumption was about 415 TWh in 2024, approximately 1.5% of global

electricity consumption, and projects continued growth as AI adoption expands [21]. The same report emphasizes both uncertainty and the importance of efficiency improvements, indicating that future energy demand depends jointly on technology, usage, and hardware trends.

The dominant architecture also creates institutional concentration: a small number of providers control much of the compute, models, deployment interfaces, and associated data. This paper investigates an alternative: a collectively operated AI infrastructure in which the model is a common computational asset and the network is open to heterogeneous contributors.

The idea is not to put neural-network training inside a blockchain. Large tensor operations are poorly matched to replicated blockchain execution. Instead, GAUD-E separates concerns. The AI layer runs off-chain across distributed nodes; a protocol layer coordinates jobs and validates results; a blockchain layer provides tamper-evident records, programmable settlement, and governance; and a community layer defines which forms of contribution are valuable.

The proposed system is inspired by several research and engineering trajectories. Federated Learning (FL) established that a shared model can be trained from decentralized data without collecting raw user data centrally [1], [2]. DiLoCo and OpenDiLoCo demonstrated low-communication distributed training across weakly connected workers [4], [5]. Gensyn has built an open protocol around off-chain ML execution, verification, peer-to-peer communication, and decentralized coordination [6], [7]. Prime Intellect has demonstrated globally distributed training and asynchronous reinforcement learning with heterogeneous contributors [9], [10]. Bittensor has implemented incentive-based subnetworks in which miners produce digital commodities and validators score their work to determine emissions [11], [12]. Akash demonstrates a decentralized marketplace for compute supply and demand [13]. These systems establish that major components of the proposed stack are technically credible; they also expose hard problems involving verification, incentives, communication, privacy, and cold-start liquidity.

This paper therefore asks a systems question: *How can an open network combine distributed AI training, verifiable contribution, privacy-preserving data use, and economic*

coordination so that the resulting intelligence is developed as a shared global resource?

II. DESIGN GOALS AND NON-GOALS

A. Design goals

The proposed network has seven primary goals:

- 1) **Collective ownership in the protocol sense:** the model, standards, and infrastructure should be stewarded through open software, transparent governance, and community participation. Legal ownership of assets or entities is a separate question and is not assumed by the protocol.
- 2) **Open participation:** heterogeneous hardware should be able to contribute where technically useful, subject to security and resource requirements.
- 3) **Privacy by default:** raw user data should remain local unless the user explicitly authorizes transfer or publication.
- 4) **Verifiable work:** contribution claims should be auditable and economically accountable.
- 5) **Economic sustainability:** rewards should ultimately be funded by genuine demand for network services and/or explicitly governed protocol emissions, not by an assumption of perpetual token appreciation.
- 6) **Energy awareness:** the scheduler should measure energy use and support carbon-aware placement instead of assuming that distributed computation is inherently greener.
- 7) **Composable infrastructure:** the protocol should allow research groups and applications to build on the same worker, validation, identity, and settlement primitives.

B. Non-goals

The proposal does not assume that blockchain is the right technology for every part of AI, that federated learning alone guarantees privacy, or that a token automatically creates economic value. It also does not claim that a globally distributed network will always be cheaper, faster, or lower-carbon than a centralized data center. These properties are empirical outcomes to be tested.

III. SYSTEM OVERVIEW

Figure 1 shows the proposed layered architecture.

A. The collective model

The protocol treats the model as a shared evolving artifact rather than an individualized model. Let M_t denote the global model at round t . A set of contributors C_t performs local training or inference-assisted optimization and submits updates Δ_i^t . A generic aggregation rule is

$$M_{t+1} = \mathcal{A}(M_t, \{\Delta_i^t, w_i^t\}_{i \in C_t}), \quad (1)$$

where $\mathcal{A}$ may be FedAvg-style aggregation [1], low-communication optimization such as DiLoCo [4], asynchronous RL aggregation, or another community-approved method.

The protocol does not require a single monolithic model. A global model can consist of a base model plus specialized experts, routing policies, adapters, retrieval indices, and evaluation suites. This allows the network to evolve different components while maintaining a common model registry.

IV. COLLECTIVE LEARNING ARCHITECTURE

A. Federated learning as a privacy-oriented starting point

Federated Learning keeps training data decentralized and aggregates local model updates [1], [2]. This is a natural primitive for GAUD-E because the project seeks to learn from many participants without requiring centralized collection of all raw data.

However, FL is not synonymous with privacy. Model updates may leak information, clients can poison the global model, and a central coordinator can become a point of failure [2], [17]. Therefore, production GAUD-E should combine FL with secure aggregation, client authentication, update clipping, anomaly detection, and potentially differential privacy where the application permits it.

Secure aggregation protocols can allow a server to recover only an aggregate update rather than an individual user's update [3]. Flower's current framework provides SecAgg and SecAgg+ implementations, demonstrating that secure-aggregation components are available in modern open-source FL tooling [22].

B. Low-communication and asynchronous training

A global network cannot assume data-center-style interconnects. DiLoCo addresses this by allowing workers to perform many local optimization steps before communicating, reducing communication requirements and improving robustness to intermittent resources [4]. OpenDiLoCo provides an open-source implementation and reported experiments across continents, making it particularly relevant to the proposed architecture [5].

Asynchronous reinforcement learning is another promising strategy. Prime Intellect's INTELLECT-2 separated inference rollout workers, verifiers, and training workers to support permissionless, heterogeneous compute and asynchronous updates [9]. Their later work has continued to optimize distributed RL at much larger model scales [10]. These results motivate a GAUD-E design in which communication-heavy pretraining is not the only training mechanism: inference-rich RL and low-communication optimization can be first-class workloads.

C. Collective model registry

Every globally accepted model version should have a cryptographically addressable manifest:

```
model_id
parent_model
weights_hash
training_manifest_hash
dataset_commitments
software_environment_hash
```

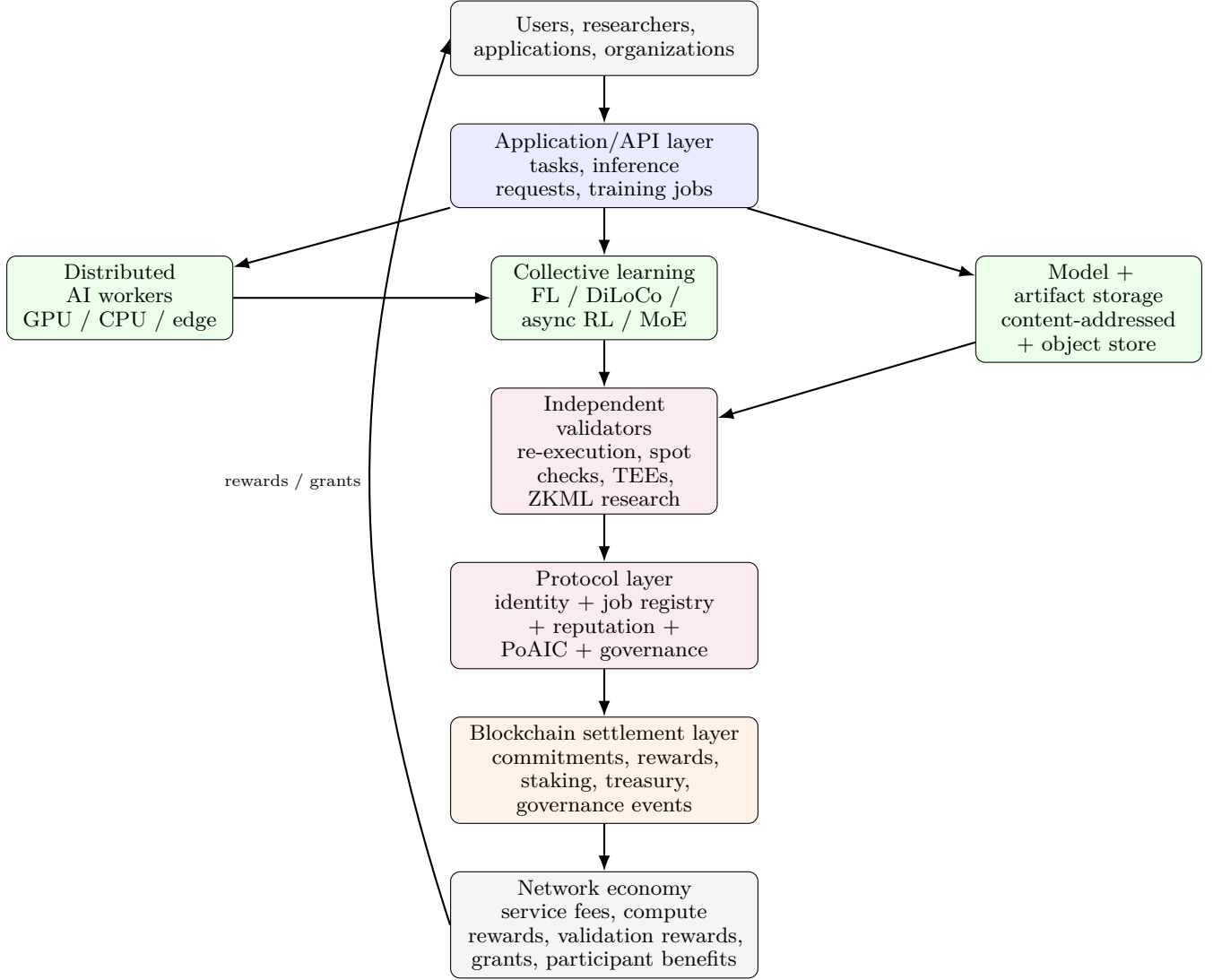


Fig. 1: Proposed GAUD-E architecture. Expensive AI computation remains off-chain; blockchain provides coordination and settlement primitives rather than executing tensor operations.

evaluation_suite_hash
contribution_set_root

The blockchain stores compact commitments to these artifacts. Large weights and datasets remain in off-chain storage. A model can therefore be traced back through a provenance graph without placing gigabytes or terabytes on-chain.

V. PROOF-OF-AI-CONTRIBUTION

A. Motivation

If contributors are economically rewarded, the network must distinguish useful work from fabricated work. This is closely related to Proof-of-Useful-Work and Proof of Learning proposals, which redirect computational effort toward machine-learning tasks [14], [15]. The problem is difficult because ML computations can be stateful, non-deterministic, heterogeneous, and expensive to reproduce.

B. Proposed PoAIC stack

We define Proof-of-AI-Contribution as a research umbrella rather than a single algorithm. A contribution record may contain:

$$R_i = H(\text{job_id} \parallel \text{worker} \parallel \text{model_hash} \parallel \text{result_hash} \parallel \text{metrics}), \quad (2)$$

plus a digital signature from the contributing worker.

A verifier can then check several increasingly strong properties:

- 1) **Execution evidence:** the node ran the specified container, model, code, and input commitments.
- 2) **Result consistency:** sampled or replicated portions of the computation match expected outputs.
- 3) **Quality:** the produced result improves or satisfies a task-specific benchmark.
- 4) **Economic integrity:** the contribution was not duplicated, self-declared, or otherwise fabricated.

- 5) **Cryptographic assurance:** where practical, TEE or zero-knowledge techniques provide stronger guarantees.

Zero-knowledge ML is an active research area. Recent surveys classify verifiable training, inference, and testing as major categories while identifying computational cost and limited coverage as continuing obstacles [16]. Gensyn also treats reproducibility and verifiable ML execution as core infrastructure problems, including machine-agnostic reproducibility tooling [8]. GAUD-E should therefore use layered verification rather than require expensive cryptographic proofs for every operation.

C. Reputation and disputes

A worker’s reward should depend on both the contribution’s measured value and its reliability history. Validators should also have reputations, and the protocol should support challenges when a validator and worker disagree. Bittensor provides a useful reference model: validators score miner outputs, and validator scores influence emissions through consensus machinery [12], [11]. GAUD-E can adapt the principle while using its own task-independent verification and economic rules.

VI. BLOCKCHAIN AND PROTOCOL LAYER

A. What belongs on-chain

The blockchain layer should store small, high-value state:

- identities or pseudonymous worker keys and role credentials;
- job commitments and contribution hashes;
- validator attestations and challenge outcomes;
- reward settlement and treasury accounting;
- governance proposals and parameter changes;
- model and dataset provenance commitments.

Training tensors, raw user data, model checkpoints, and large evaluation traces remain off-chain. This architectural boundary is essential for cost and scalability.

B. Token role

The proposed native asset, **GAUDE**, should initially be treated as a protocol utility and accounting instrument. Potential uses include service payments, compute rewards, validation rewards, staking, grants, and governance. The token should not be described as a guaranteed investment or dividend-bearing security. Whether any public issuance or secondary-market activity is lawful depends on jurisdiction-specific facts and the rights attached to the asset; those issues require legal review and are outside the technical scope of this paper.

C. Economic settlement

The protocol can support two complementary flows:

- 1) **market revenue:** users or organizations pay for inference, training, evaluation, or compute;

- 2) **protocol incentives:** an explicitly governed token-emission or grant mechanism bootstraps participation when real demand is insufficient.

The long-term target is for participant rewards to be increasingly supported by genuine network usage. This avoids an economic design that depends on continuous speculative inflows.

VII. HOW PARTICIPANTS BENEFIT

The phrase “global AI for everyone and by everyone” has four distinct meanings in this architecture.

A. Access benefit

Participants can use the collective AI without needing to own the full training infrastructure. This lowers the capital barrier to access, although service fees, latency, and hardware constraints still exist.

B. Contribution benefit

Participants who supply useful compute, validation, engineering, research, or other authorized resources can accumulate contribution reputation and, when the protocol is economically active, receive rewards.

C. Governance benefit

Participants can propose and vote on protocol parameters, benchmarks, model-release rules, treasury allocations, and other public-good decisions under a governance design that limits capture by pure token wealth where appropriate.

D. Ecosystem benefit

Researchers and developers gain an open substrate for experimenting with models and distributed training. New applications can build on the same infrastructure rather than each recreating a private compute and verification stack.

The network should distinguish *contribution accounting* from *legal ownership*. A high contribution score does not automatically create a legal equity claim in a company or foundation. The protocol can, however, make contribution and governance rules explicit and auditable.

VIII. PRIVACY AND DATA GOVERNANCE

A key design question is how a collective AI can learn from its participants without turning the system into a global raw-data warehouse.

The default pipeline should be:

- 1) data remains on the user’s device or organization-controlled environment;
- 2) the user or data controller explicitly opts into a training task;
- 3) the local worker computes an update or task result;
- 4) secure aggregation masks individual updates where the task supports it;

- 5) the network retains only the minimum metadata needed for verification, accounting, and model provenance.

This is stronger than simply saying “data is private.” Federated Learning reduces central collection, but privacy leakage can still occur through updates and inference attacks [2], [17]. Therefore, GAUD-E should implement a formal data-governance layer covering consent, purpose limitation, retention, deletion, access control, and provenance.

For user-provided datasets that are intentionally shared with the network, the system should record license and provenance metadata. Data that cannot legally or ethically be used for collaborative training should not enter the shared model pipeline.

IX. ENERGY, COST, AND CARBON HYPOTHESES

It is tempting to argue that distributed AI is automatically cheaper or greener because it reuses idle hardware. The literature does not justify that universal claim. The energy profile depends on hardware efficiency, utilization, network traffic, workload characteristics, electricity mix, and the baseline being compared.

The IEA projects strong growth in data-center electricity demand and notes that AI is a major driver of accelerated-server growth [21]. Research measuring AI inference also finds substantial variation in energy cost across model types and task formulations [20]. Federated learning can reduce data movement, but communication itself can be a significant energy component, motivating sparsification and client-selection techniques [18], [19].

GAUD-E should therefore measure environmental impact explicitly:

$$E_{job} = E_{compute} + E_{network} + E_{storage} + E_{overhead}, \quad (3)$$

$$CO_2e_{job} = \sum_r E_{job,r} CI_r, \quad (4)$$

where CI_r is the carbon intensity of the electricity used in region r . The scheduler can then choose between minimizing cost, latency, energy, or carbon according to the task objective.

The core hypothesis is therefore not “blockchain reduces emissions,” but:

A distributed AI network can reduce total cost or carbon for some workloads if it achieves high utilization of existing resources, limits communication overhead, and routes work toward efficient hardware and lower-carbon electricity.

This is a testable claim.

X. THREAT MODEL

The system must assume that some participants are malicious or unreliable. Relevant threats include:

- fake work claims and replayed receipts;
- model poisoning and targeted backdoors;
- Sybil attacks through many identities;
- collusion between workers and validators;

TABLE I: Reference technology stack for the GAUD-E prototype.

Layer	Candidate technologies / role
AI framework	PyTorch; model-specific runtimes such as vLLM where appropriate
Federated learning	FedAvg baseline; secure aggregation; later DiLoCo / async RL
P2P	libp2p, DHT, gossip, authenticated channels
Worker	Python service; hardware discovery; containerized workloads
Storage	Object storage plus content-addressed artifacts such as IPFS-style addressing
Validation	Re-execution, spot checks, benchmark scoring, TEE and ZKML research
Protocol API	FastAPI / gRPC-style service boundary
Blockchain	EVM-compatible chain for settlement and governance
Contracts	Solidity with audited standard libraries such as OpenZeppelin
Observability	Prometheus / Grafana; job-level energy and carbon metrics
Frontend	Web dashboard, explorer, worker status, governance

- privacy leakage from updates or model outputs;
- denial-of-service and network-level resource exhaustion;
- compromised worker machines;
- smart-contract vulnerabilities and key compromise;
- economic manipulation of task prices or token incentives.

Recent blockchain-FL reviews explicitly identify model poisoning, backdoors, Sybil attacks, and inference attacks as important threats [17]. GAUD-E should therefore separate identity, reputation, stake, and task performance rather than rely on any single signal.

A practical roadmap is:

signed receipts → replication → independent validation → challenge → (5)

XI. TECHNICAL STACK

Table I summarizes a plausible implementation stack. The table is deliberately framed as a reference architecture rather than a fixed dependency list.

XII. REFERENCE ARCHITECTURE FOR THE FIRST RELEASE

A practical first release should not attempt to solve every decentralized-systems problem simultaneously. A staged architecture is safer.

A. M0: Reproducible local experiment

Run a single coordinator and multiple local workers. Demonstrate deterministic job manifests, local model training, contribution receipts, and reproducible evaluation.

TABLE II: Proposed evaluation questions and metrics.

Question	Example metrics
Learning quality	validation loss, task accuracy, benchmark score, convergence time
Communication	GB transferred per round, update frequency, retransmissions
Verification	verifier cost, false-accept rate, false-reject rate, audit coverage
Reliability	node dropout tolerance, recovery time, stale-update impact
Security	poisoning success rate, Sybil resistance, challenge detection rate
Economics	cost per useful training step, reward concentration, revenue per GPU-hour
Energy	kWh per job, kWh per useful token/step, kgCO ₂ e per job
Fairness	reward Gini / concentration, small-device participation rate
Privacy	leakage under defined attack models, client anonymity properties

B. M1: Federated collective-learning network

Move from local simulation to real multi-machine training. Add secure aggregation, model checkpoints, client authentication, and a model registry.

C. M2: Verification network

Add independent validators, randomized audits, reputation, challenges, and explicit reward settlement.

D. M3: P2P network

Replace the coordinator’s discovery and scheduling role with permissionless peer discovery and decentralized task routing while preserving the chain as a compact settlement layer.

E. M4: Collective foundation model

Scale the training system using low-communication optimization and asynchronous RL. Publish model cards, datasets or dataset commitments, evaluation results, and complete provenance.

F. M5: Public economy

Only after the technical network has useful demand should the token economy move beyond testnet experimentation. Public issuance, governance rights, or value-sharing mechanisms should be reviewed against applicable regulation before deployment.

XIII. EVALUATION METHODOLOGY

A scientific implementation should be evaluated against centralized and simpler decentralized baselines. The proposed benchmark suite includes:

For economic analysis, the correct baseline is not “zero cost.” The comparison should include cloud GPU rental, local hardware depreciation, networking, storage, validation overhead, and protocol fees. For carbon analysis, the comparison should include embodied impacts when data are available, in addition to operational electricity.

TABLE III: Selected related systems and the design lessons they provide.

Reference	Demonstrated idea	Implication for GAUDE
Federated Learning	Shared model training while raw data remain decentralized	Use as the privacy-oriented baseline; add verification and open participation
DiLoCo / OpenDiLoCo	Low-communication distributed training across weakly connected workers	Use local-update and asynchronous patterns when internet bandwidth is limited
Gensyn	Decentralized ML execution, verification, P2P communication, and coordination	Reuse the problem decomposition: off-chain compute with protocol-level verification and settlement
Prime Intellect	Globally distributed training and asynchronous RL with heterogeneous compute	Favor RL and other low-communication workloads for early global experiments
Bittensor	Miners produce AI-related commodities; validators score work to drive emissions	Use contribution scoring and validator incentives, but orient the primary output toward a shared evolving model
Akash	Decentralized compute marketplace	Treat compute as an open resource layer above which AI-specific verification and model evolution can be built
Proof-of-Useful-Work / Proof of Learning	Redirect blockchain computation toward useful ML work	Provides conceptual precedent for linking useful AI computation to consensus or economic incentives

XIV. RELATED SYSTEMS AND POSITIONING

Table III summarizes the closest reference directions.

The positioning is therefore best understood as a systems synthesis: the proposed network combines the privacy model of collaborative learning, the communication patterns of distributed training, the verification problem of decentralized ML, and the incentive logic of compute/AI networks.

XV. GOVERNANCE AND COLLECTIVE STEWARDSHIP

A global collective AI needs more than a blockchain. It needs rules for model releases, benchmark changes, harmful-task restrictions, treasury spending, software upgrades, and conflict resolution.

A proposed governance structure has three layers:

- 1) **Protocol governance:** token holders and/or reputation-qualified contributors vote on technical parameters.
- 2) **Technical councils:** elected or rotating maintainers review security-critical changes.
- 3) **Public research process:** model specifications, benchmark proposals, and major training changes are documented through open RFCs.

Governance should avoid pure plutocracy where one participant can buy all influence. Hybrid mechanisms can weight long-term contribution, reputation, and stake differently for different decision types. Any such mechanism must be empirically tested for capture and collusion.

XVI. LIMITATIONS AND OPEN RESEARCH PROBLEMS

Several challenges remain unsolved or only partially solved:

- 1) **Verification cost:** strong verification of large ML jobs can itself require substantial compute [7], [16].
- 2) **Heterogeneous hardware:** reproducibility and numerical differences across GPUs complicate direct comparison.
- 3) **Communication bottlenecks:** internet-scale training is not equivalent to intra-cluster training [4].
- 4) **Data poisoning:** an open system can accept useful as well as malicious updates.
- 5) **Sybil resistance:** open identity without defenses can distort rewards and governance.
- 6) **Cold-start economics:** compute supply and AI demand must grow together; decentralized marketplaces face a bootstrapping problem [7].
- 7) **Regulation and rights:** token rights, user-data law, model licenses, and cross-border processing need jurisdiction-specific analysis.
- 8) **Environmental accounting:** distributed use of consumer hardware is not automatically lower-carbon than optimized data centers.
- 9) **Model quality and alignment:** collective training can improve a model but can also magnify noisy or adversarial objectives.

XVII. CONCLUSION

This paper proposes GAUD-E Collective Network as an open architecture for a global collective artificial intelligence. The core idea is simple: many participants contribute useful resources to a shared AI system, the network verifies those contributions, the blockchain records compact commitments and economic settlements, and the resulting intelligence is continuously improved as a collective artifact.

The proposal does not rely on a claim that blockchain makes AI cheaper by itself. Instead, it uses blockchain only where a blockchain is useful: identity, commitments, verification outcomes, settlement, staking, and governance. The AI work remains off-chain and can use federated learning, low-communication optimization, asynchronous reinforcement learning, and other distributed-training techniques.

Existing systems demonstrate the feasibility of many individual components. Federated learning shows that shared models can be trained without centralizing raw data; DiLoCo and OpenDiLoCo show how training can tolerate weaker communication; Gensyn demonstrates protocol-level approaches to decentralized ML execution and verification; Prime Intellect demonstrates globally distributed asynchronous RL; Bittensor demonstrates incentive mechanisms around valuable AI-related work; and decentralized cloud networks such as Akash demonstrate distributed compute supply. The research challenge for GAUD-E is to integrate these directions into a coherent system whose primary output is a continuously improving collective model and whose economics reward demonstrably useful participation.

The next scientific milestone is therefore not a speculative token launch. It is an empirical demonstration: independent machines should be able to participate in the same training experiment, produce verifiable contributions, update a common model, and settle rewards transparently. If that experiment succeeds under measured security, cost, privacy, and energy constraints, the project will have a credible foundation for further scaling.

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This manuscript is a conceptual research document for the open-source GAUD-E Collective Network project. It distinguishes proposed mechanisms from established capabilities and does not present unpublished performance results as empirical facts.

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